# § 10.5 Fourier Series

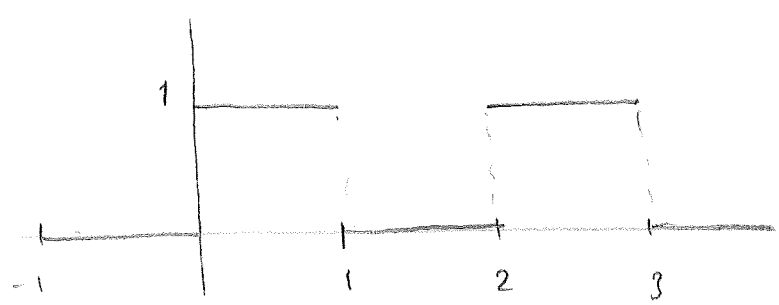
With Taylor series we approximated a fn using powers of $(x-a)$, $(x-a)^n$.

For Fourier series, we approximate a fn using sines and cosines of the type $\sin kx$, $\cos kx$.

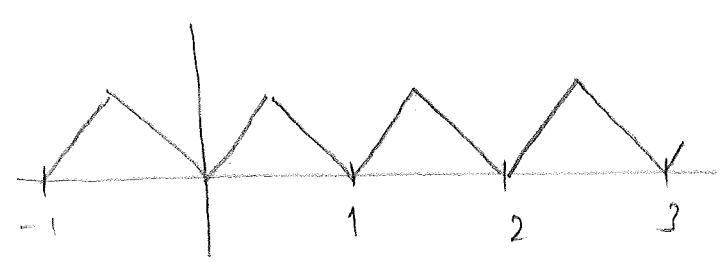
Since these fns are all periodic, we can only hope to approximate periodic fns.

e.g.

Square Wave




Triangular Wave



We will attempt to approximate $f$ with a sum of trig. fns of the form

$$f(x) \approx F_n(x)$$

$$= a_0 + a_1 \cos x + a_2 \cos(2x) + a_3 \cos(3x) + \cdots + a_n \cos(nx)$$

$$+ b_1 \sin x + b_2 \sin(2x) + b_3 \sin(3x) + \cdots + b_n \sin(nx).$$

$F_n(x)$ is called the <u>Fourier polynomial of degree $n$</u> (even though it isn't actually a polynomial) and the numbers $a_k$, $b_k$ are called the <u>Fourier coefficients</u>.

Note that since all the fns $\cos(kx)$, $\sin(kx)$ repeat every $2\pi$, so must the original fn $f(x)$.

For such a fn there is a formula for the Fourier coefficients which gives a good approximation to $f$.

# The Fourier Coefficients for a Periodic Function $f$ of Period $2\pi$

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x)\,dx$$

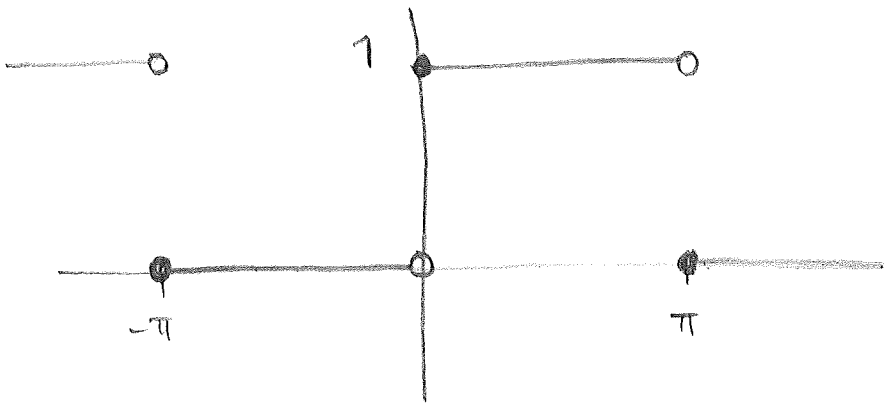
$$a_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)\cos(kx)\,dx, \qquad k > 0$$

$$b_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)\sin(kx)\,dx, \qquad k > 0$$

Note that $a_0$ is simply the average value of $f$ on $[-\pi, \pi]$.

Ex. Construct successive Fourier polynomials for the square wave fn $f$ of period $2\pi$ given by

$$f(x) = \begin{cases} 0, & -\pi \leq x < 0 \\ 1, & 0 \leq x < \pi \end{cases}$$




Since $a_0$ is the average value on $[-\pi, \pi]$, it's reasonable to guess that $a_0 = \frac{1}{2}$. Checking gives

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x)\,dx$$

$$= \frac{1}{2\pi} \int_{-\pi}^{0} 0\,dx + \frac{1}{2\pi} \int_{0}^{\pi} 1\,dx = 0 + \frac{1}{2\pi}(\pi) = \frac{1}{2}.$$

$$a_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(kx)\,dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{0} 0 \, dx + \frac{1}{2\pi} \int_{0}^{\pi} \cos(kx)\,dx$$

$$= 0 + \frac{1}{\pi} \left[ \frac{\sin(kx)}{k} \right]_0^{\pi}$$

$$= 0 + \frac{1}{\pi} \left( \frac{\sin(k\pi)}{k} - \frac{\sin 0}{k} \right)$$

$$= 0.$$

$$b_k = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(kx)\,dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{0} 0 \, dx + \frac{1}{\pi} \int_{0}^{\pi} \sin(kx)\,dx.$$

$$= 0 + \frac{1}{\pi} \left[ -\frac{1}{k} \cos(kx) \right]_0^{\pi}$$

$$= \frac{1}{\pi} \left( -\frac{1}{k} \cos(k\pi) - \left( -\frac{1}{k} \cos 0 \right) \right)$$

$$= \frac{1}{\pi k}\left(1 - \cos(k\pi)\right)$$

Now $\cos(k\pi) = \begin{cases} -1, & k \text{ odd} \\ 1, & k \text{ even} \end{cases}$

and so

$$b_k = \begin{cases} \frac{2}{\pi k}, & k \text{ odd} \\ 0, & k \text{ even.} \end{cases}$$

From this we see that

$$F_1(x) = \frac{1}{2} + \frac{2}{\pi}\sin x$$

$$F_2(x) = \frac{1}{2} + \frac{2}{\pi}\sin x \qquad (+\ 0\sin(2x))$$

$$F_3(x) = \frac{1}{2} + \frac{2}{\pi}\sin x + \frac{2}{3\pi}\sin(3x)$$

$$F_4(x) = \frac{1}{2} + \frac{2}{\pi}\sin x + \frac{2}{3\pi}\sin(3x) \qquad (+\ 0\sin(4x))$$

$$F_5(x) = \frac{1}{2} + \frac{2}{\pi}\sin x + \frac{2}{3\pi}\sin(3x) + \frac{2}{5\pi}\sin(5x).$$

etc.

Using Maple, we can see how the Fourier polynomials give better and better approximations to the square wave.

Note that as $n \to \infty$, the approximation given by $F_n$ gets better everywhere (except for the points of discontinuity).

Note also that this is different to the situation with Taylor polynomials where the approximation is usually just on an interval (ie. locally).

As $n \to \infty$, the Fourier polynomials $F_n$ give us the <u>Fourier Series</u> for $f$.

On the interval $[-\pi, \pi]$, the Fourier Series for $f$ is given by

$$a_0 + a_1 \cos x + a_2 \cos(2x) + a_3 \cos(3x) + \cdots$$
$$+ b_1 \sin x + b_2 \sin(2x) + b_3 \sin(3x) + \cdots$$

where $a_k$ and $b_k$ are the Fourier coefficients whose formulae were given earlier.

If $f$ is differentiable, then this Fourier series converges to $f(x)$ and we have

$$f(x) = a_0 + a_1 \cos x + a_2 \cos(2x) + a_3 \cos(3x) + \cdots$$
$$+ b_1 \sin x + b_2 \sin(2x) + b_3 \sin(3x) + \cdots$$

The term

$$a_1 \cos x + b_1 \sin x$$

of period $2\pi$ is called the <u>fundamental harmonic</u> of $f$ while the term

$$a_k \cos(kx) + b_k \sin(kx)$$

of period $\frac{2\pi}{k}$ is called the <u>kth harmonic</u> of $f$.

e.g In the last example, the fundamental harmonic was

$$\frac{2}{\pi} \sin x$$

while the k-th harmonic was

$$\frac{2}{k\pi} \sin(kx), \qquad k \text{ odd}$$

$$0, \qquad k \text{ even}$$

# Energy and the Energy Theorem

We define the energy $E$ of a periodic fn $f$ of period $2\pi$ by

$$E = \frac{1}{\pi}\int_{-\pi}^{\pi} (f(x))^2\,dx.$$

One can check that for the kth harmonic

$$a_k \cos(kx) + b_k \sin(kx)$$

the energy is given by

$$\frac{1}{\pi}\int_{-\pi}^{\pi} (a_k \cos(kx) + b_k \sin(kx))^2\,dx = a_k^2 + b_k^2$$

$$= A_k^2$$

where $A_k = \sqrt{a_k^2 + b_k^2}$ is called the <u>amplitude</u> of the k-th harmonic.

The energy of the constant term $a_0$ in the Fourier series is given by

$$E = \frac{1}{\pi}\int_{-\pi}^{\pi} a_0^2 \, dx = 2a_0^2$$

and so we define the amplitude $A_0$ of the constant term by

$$A_0 = \sqrt{2A_0^2} = \sqrt{2}\cdot a_0.$$

The big result is then that for a fn $f$ which is periodic of period $2\pi$

$$E = \frac{1}{\pi}\int_{-\pi}^{\pi} (f(x))^2 \, dx = A_0^2 + A_1^2 + A_2^2 + A_3^2 + \cdots$$

$$= 2a_0^2 + a_1^2 + a_2^2 + a_3^2 + \cdots$$
$$+ b_1^2 + b_2^2 + b_3^2 + \cdots$$

This is the so-called energy theorem (also known as Parseval's identity).

It can be thought of as an infinite-dimensional version of Pythagoras' theorem.

e.g. In the example with the square wave

$$E = \frac{1}{\pi}\int_{-\pi}^{\pi} (f(x))^2\,dx = \frac{1}{\pi}\int_{0}^{\pi} 1^2\,dx = 1.$$

$$a_0 = \frac{1}{2}, \quad a_k = 0, \quad k \geq 1$$

$$b_k = \begin{cases} \frac{2\pi}{k}, & k \text{ odd} \\ 0, & k \text{ even} \end{cases}$$

Thus $A_0^2 = 2a_0^2 = 2\left(\frac{1}{2}\right)^2 = \frac{1}{2}.$

$$A_k^2 = \begin{cases} \dfrac{4}{\pi^2 k^2}, & k \text{ odd} \\ 0, & k \text{ even.} \end{cases}$$

Then

$$A_0^2 + A_1^2 + A_2^2 + A_3^2 + \cdots$$

$$= \frac{1}{2} + \frac{4}{\pi^2 1^2} + 0 + \frac{4}{\pi^2 \cdot 3^2} + \cdots$$

$$= \frac{1}{2} + \frac{4}{\pi^2}\left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots\right)$$

$$= \frac{1}{2} + \frac{4}{\pi^2}\left(\frac{\pi^2}{8}\right)$$

$$= \frac{1}{2} + \frac{1}{2} = 1$$ using the fact that

$$1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \cdots = \frac{\pi^2}{8}.$$

# Functions which Have Periods Other than $2\pi$

Suppose $f(x)$ has period $b$.

If we let

$$x = \frac{bt}{2\pi},$$

then as $x$ ranges over the interval $[-b/2, b/2]$, $t$ will range over the interval $[-\pi, \pi]$.

Now set

$$g(t) = f\left(\frac{bt}{2\pi}\right) = f(x).$$

$g$ is then periodic of period $2\pi$ and we can find the Fourier series for $g$ in the usual way.

$$g(t) = a_0 + a_1 \cos t + a_2 \cos(2t) + a_3 \cos(3t) + \cdots$$
$$+ b_1 \sin t + b_2 \sin(2t) + b_3 \sin(3t) + \cdots$$

Thus, since $t = \frac{2\pi x}{b}$

$$f(x) = g(t) = a_0 + a_1 \cos\left(\frac{2\pi x}{b}\right) + a_2 \cos\left(\frac{4\pi x}{b}\right) + a_3 \cos\left(\frac{6\pi x}{b}\right) + \cdots$$
$$+ b_1 \sin\left(\frac{2\pi x}{b}\right) + b_2 \sin\left(\frac{4\pi x}{b}\right) + b_3 \sin\left(\frac{6\pi x}{b}\right) + \cdots$$

What about the Fourier coefficients?

Well.

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} g(t)\, dt = \frac{1}{2\pi} \int_{-b/2}^{b/2} f(x) \cdot \frac{2\pi}{b}\, dx$$
$$= \frac{1}{b} \int_{-b/2}^{b/2} f(x)\, dx$$

as $g(t) = f(x)$ and $t = \frac{2\pi x}{b}$

so $dt = \frac{2\pi}{b}\, dx$.

For $k \geq 1$

$$a_k = \frac{1}{\pi} \int_{-\pi}^{\pi} g(t) \cos(kt)\, dt$$

$$= \frac{1}{\pi} \int_{-b/2}^{b/2} f(x) \cdot \cos\left(\frac{2\pi k x}{b}\right) \cdot \frac{2\pi\, dx}{b}$$

again, $dt = \frac{2\pi\, dx}{b}$

$$= \frac{2}{b} \int_{-b/2}^{b/2} f(x) \cos\left(\frac{2\pi k x}{b}\right) dx.$$

Similarly, for $k \geq 1$.

$$b_k = \frac{2}{b} \int_{-b/2}^{b/2} f(x) \sin\left(\frac{2\pi k x}{b}\right) dx.$$

To summarize.

If $f$ is periodic of period $b$, then the Fourier series for $f$ on $[-b/2, b/2]$ is given by

$$f(x) = a_0 + \sum_{k=1}^{\infty} \left( a_k \cos\left(\frac{2\pi k x}{b}\right) + b_k \sin\left(\frac{2\pi k x}{b}\right)\right)$$

where

$$a_0 = \frac{1}{b} \int_{-\frac{b}{2}}^{\frac{b}{2}} f(x)\,dx,$$

and for $k \geq 1$,

$$a_k = \frac{2}{b} \int_{-b/2}^{b/2} f(x) \cos\left(\frac{2\pi k x}{b}\right) dx,$$

$$b_k = \frac{2}{b} \int_{-b/2}^{b/2} f(x) \sin\left(\frac{2\pi k x}{b}\right) dx.$$

Ex. $f(x) = x^2$ on $[-1, 1]$

Here $b = 2$, $b/2 = 1$ and

$$a_0 = \frac{1}{2}\int_{-1}^{1} x^2\,dx = \frac{1}{2}\left[\frac{x^3}{3}\right]_{-1}^{1} = \frac{1}{2}\left(\frac{1}{3} - \left(-\frac{1}{3}\right)\right) = \frac{1}{3}.$$

For $k \geq 1$

$$a_k = \frac{2}{2}\int_{-1}^{1} x^2 \cos\left(\frac{2\pi k x}{2}\right) dx$$

$$= \int_{-1}^{1} x^2 \cos(\pi k x)\,dx$$

by parts

$u = x^2$, $dv = \cos(\pi k x)\,dx$

$du = 2x\,dx$, $v = \frac{\sin(\pi k x)}{\pi k}$

$$= \left[\frac{x^2 \sin(\pi k x)}{\pi k}\right]_{-1}^{1} - \frac{2}{\pi k}\int_{-1}^{1} x \sin(\pi k x)\,dx$$

by parts again

$u = x$, $dv = \sin(\pi k x)\,dx$

$du = dx$, $v = -\frac{\cos(\pi k x)}{\pi k}$

$$= \left[\frac{x^2 \sin(\pi k x)}{\pi k}\right]_{-1}^{1} - \frac{2}{\pi k}\left\{\left[\frac{-x\cos(\pi k x)}{\pi k}\right]_{-1}^{1} - \int_{-1}^{1} \frac{-\cos(\pi k x)}{\pi k}\,dx\right\}$$

$$= \left[ \frac{x^2 \sin(\pi k x)}{\pi k} \right]_{-1}^{1} - \frac{2}{\pi k} \left\{ \left[ \frac{-x \cos(\pi k x)}{\pi k} \right]_{-1}^{1} + \int_{-1}^{1} \frac{\cos(\pi k x)}{\pi k} \, dx \right\}$$

$$= \left[ \frac{x^2 \sin(\pi k x)}{\pi k} \right]_{-1}^{1} - \frac{2}{\pi k} \left\{ \left[ \frac{-x \cos(\pi k x)}{\pi k} \right]_{-1}^{1} + \left[ \frac{\sin(\pi k x)}{\pi^2 k^2} \right]_{-1}^{1} \right\}$$

$$= \frac{1^2 \sin(\pi k)}{\pi k} - \frac{(-1)^2 \sin(-\pi k)}{\pi k}$$

$$- \frac{2}{\pi k} \left\{ \frac{-1 \cos(\pi k)}{\pi k} - \left( \frac{-(1) \cos(-\pi k)}{\pi k} \right) + \frac{\sin(\pi k)}{\pi^2 k^2} - \frac{\sin(-\pi k)}{\pi^2 k^2} \right\}$$

Now $\sin(\pi k) = \sin(-\pi k) = 0$

while $\cos(\pi k) = \cos(-\pi k) = (-1)^k$

and so we get

$$a_k = 0 - 0$$

$$- \frac{2}{\pi k} \left\{ - \frac{2(-1)^k}{\pi k} + 0 - 0 \right\}$$

$$= \frac{4}{\pi^2 k^2} (-1)^k.$$

Now

$$b_k = \frac{2}{2}\int_{-1}^{1} x^2 \sin(\pi k x)\,dx.$$

We could integrate this by parts as before, but it is much simpler to observe that as the integrand is an <u>odd</u> fn and we are integrating over an interval which is symmetric about 0, we must have

$$b_k = 0.$$

Thus we have

$$x^2 = \frac{1}{3} + \frac{4}{\pi^2}\sum_{k=1}^{\infty} \frac{(-1)^k}{k^2}\cos(\pi k x), \qquad -1 \le x \le 1.$$

Note that in this example which was an even fn, there were no sine terms in the Fourier series while for the last example (the square wave), there were no cosine terms.

This is an example of the following general feature

- The Fourier series on $[-b/2, b/2]$ for a fn which is <u>even</u> contains no sine terms.

- The Fourier series on $[-b/2, b/2]$ for a fn which is either <u>odd</u> or a vertically shifted <u>odd</u> fn contains no cosine terms.

Finally, some entertainment!

If we let $x = 1$ in

$$x^2 = \frac{1}{3} + \frac{4}{\pi^2} \sum_{k=1}^{\infty} \frac{(-1)^k}{k^2} \cos(\pi k x)$$

we get

$$1 = \frac{1}{3} + \frac{4}{\pi^2} \sum_{k=1}^{\infty} \frac{(-1)^k}{k^2} \cos(\pi k)$$

and since $\cos(\pi k) = (-1)^k$

$$1 = \frac{1}{3} + \frac{4}{\pi^2} \sum_{k=1}^{\infty} \frac{(-1)^k}{k^2} (-1)^k$$

$$1 = \frac{1}{3} + \frac{4}{\pi^2} \sum_{k=1}^{\infty} \frac{(-1)^{2k}}{k^2}$$

So

$$1 = \frac{1}{3} + \frac{4}{\pi^2} \sum_{k=1}^{\infty} \frac{1}{k^2},$$

$$\frac{2}{3} = \frac{4}{\pi^2} \sum_{k=1}^{\infty} \frac{1}{k^2}$$

Thus

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{2}{3} \cdot \frac{\pi^2}{4} = \frac{\pi^2}{6}$$

So using Fourier series we have proved that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6} \quad !$$